

Q1. Illustrate the problem faced by Oligopolistic firms with help of the Prisoners Dilemma?

Oligopolistic firms operate in markets with few competitors, which creates a unique challenge in their decision-making. They must balance cooperation (to maintain stable profits) with competition (to maximize individual gains). The **Prisoner's Dilemma** is often used to illustrate this problem.

The Problem:

- **Scenario:** Two firms, A and B, dominate a market. Each firm can either **cooperate** (not cut prices) or **defect** (cut prices to gain more market share).
- **Payoff Matrix:**

Firm B	Cooperate (Don't Cut Prices)	Defect (Cut Prices)
Firm A Cooperates	High Profits for Both (A: \$10M, B: \$10M)	A Loses, B Gains (A: \$5M, B: \$15M)
Firm A Defects	B Loses, A Gains (A: \$15M, B: \$5M)	Low Profits for Both (A: \$7M, B: \$7M)

Explanation:

1. **If both firms cooperate** (don't cut prices), they maximize joint profits (e.g., \$10M each).
2. **If one defects** (cuts prices) while the other cooperates, the defector gains a larger share, while the cooperator suffers losses.
3. **If both defects**, they engage in a price war, reducing profits for both (e.g., \$7M each).

Dilemma:

- From an individual perspective, each firm has an incentive to defect since it could increase their profit (if the other cooperates).
- However, if both defects, they end up worse off than if they had cooperated.

Key Insight:

The **Prisoner's Dilemma** highlights the conflict between self-interest and collective interest. In oligopolies:

- Firms benefit from implicit or explicit cooperation (e.g., price-fixing agreements).
- Yet, due to fear of being undercut by rivals, they may choose competitive strategies that harm all players.

Q2. Explain Education as a Signaling Model?

The signaling model of education is an economic theory that explains how education can signal a worker's ability to potential employers. The model suggests that education is a way for workers to communicate their skills to employers, rather than a way to improve their skills.

Explanation

Signaling theory

This theory was first proposed by Michael Spence in 1973. It's based on the idea that markets have asymmetric information, meaning that buyers and sellers don't have the same information.

Education as a signal

In the job market, employees have more information about their skills and education than employers. Education can signal to employers that a worker is more capable, and therefore more likely to be productive.

Screening

The signaling model of education screens out low-productivity workers and allows high-productivity workers to move into higher-paying positions.

Credential

The model suggests that education provides workers with a credential that employers can use to judge their capabilities.

Limitations

Some critics argue that the signaling model suggests that education doesn't improve a society's economic productivity.

Example

A college degree tells bosses that a person is smart and works hard. Bosses use it to choose good workers.

Q3. Explain the Stackelberg's Model of Duopoly?

The Stackelberg model of duopoly is a microeconomic model that describes the strategic interaction between two firms in a market. In this model, one firm is the leader and the other is the follower. This model was introduced by **Heinrich von Stackelberg** in 1934.

Key features

The **leader** decides how much to produce first.

The **follower** waits, sees the leader's choice, and then decides its own production.

Sequential moves

The Stackelberg model is a sequential move game, unlike the Cournot model where firms choose simultaneously.

Strategic advantage

The leader firm has a strategic advantage because they set the production quantity first.

Goal

The goal of both firms is to maximize profit while competing with each other.

Nash equilibrium

The desired outcome is for the firms to converge to a Nash equilibrium, where all players are in their best situation.

Real-world applications

- The Stackelberg model can be used to analyze the dynamics of real-world market.
- The model can be used to analyze the impact of changing model parameters on the market dynamics.

Related concepts

- The Stackelberg model is a type of oligopoly market model.
- The Stackelberg model is also known as the "leader-follower game".

Example:

Imagine two companies selling the same type of phone.

- **Company A (leader)** decides to make 1,000 phones.
 - **Company B (follower)** sees this and decides to make 800 phones because making more would lower the price too much.
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Advance Microeconomics

Past Paper-Fall (2023)

ECON-314

Q1. Why does price leadership sometimes evolve in oligopolistic markets? Explain how the price leader determines a profit maximizing price.

Price leadership evolves in oligopolistic markets because of market interdependence and the desire to balance profit maximization with market share.

Explanation

Market interdependence

In oligopolistic markets, a small number of firms dominate the industry, and their actions affect each other.

Profit maximization

The price leader sets a price that maximizes its profit, taking into account its costs and how competitors might react.

Market share

The price leader also considers how its price might affect its market share.

How the price leader determines a profit maximizing price

Factors

The price leader considers factors like cost changes, demand fluctuations, and competitive pressures.

Goal

The price leader's goal is to balance profit maximization with market share.

Price stability

The price leader's strategy can help maintain price stability and reduce price rivalry.

Types of price leadership

Dominant firm price leadership: A single, highly influential firm sets the price standard for the entire market.

Collusive price leadership: A handful of dominant firms agree to keep their prices in alignment.

Q2. Discuss the Inefficiency of Competition with Externalities?

Competition with externalities is inefficient because it leads to nonoptimal production levels. This is due to the fact that individuals and firms do not pay for the costs or receive the benefits of their actions on others.

Explanation

Negative externalities

When a firm or individual's actions cause harm to others, but they don't pay for it. This leads to overproduction of goods with negative externalities.

Positive externalities

When a firm or individual's actions benefit others, but they don't receive compensation for it. This leads to underproduction of goods with positive externalities.

Market inefficiency

The wedge between social and private costs or returns creates inefficient market outcomes.

Government intervention

Governments often intervene to correct externalities and ensure that costs and benefits are fully internalized.

Examples

Air pollution

Burning coal creates air pollution that harms others, but the polluter doesn't pay for it.

Groundwater pollution

Using fertilizer creates groundwater pollution that harms others, but the polluter doesn't pay for it.

Noise pollution

Listening to loud music impacts others' ability to enjoy a quiet night, but the listener doesn't pay for it.

Solutions

Government intervention

Governments can create policies to correct externalities, such as cap-and-trade systems.

Market-based solutions

Markets can be used to correct externalities, but government intervention may still be required.

Q1. Discuss Nash equilibrium in respect of Prisoners Dilemma?

In game theory, the Nash equilibrium is a decision-making theorem that describes when a player's best chance at a desired outcome is to stick with their initial strategy. The prisoner's dilemma is a well-known example of a Nash equilibrium.

Explanation

- In the prisoner's dilemma, two criminals are interrogated separately for the same crime.
- The Nash equilibrium in this case is for both prisoners to confess, even though it leads to a suboptimal outcome for both.
- The dominant strategy in the prisoner's dilemma is to betray, meaning that confessing always gives a higher payoff regardless of what the other player chooses.

- However, if both prisoners cooperated and remained silent, they would both receive a better outcome.
- The problem with cooperating is that it's difficult to trust the other prisoner to do the same.

Significance

- The Nash equilibrium is a key concept in game theory.
- The Nash equilibrium is not always the most optimal strategy.

Q2. Explain Global Risk Aversion and measure of relative risk aversion?

Global risk aversion is a measure of how risk-averse people are on a global scale. It's a significant factor in business cycles and international equity correlations.

Explanation

Risk aversion: The tendency to prefer certainty over uncertainty. Risk-averse people avoid risky situations and prefer low-risk investments.

Global risk aversion: A measure of the average curvature of utility functions across the world.

Relative risk aversion: A measure of how risk-averse a decision maker is. It depends on the outcome variable used in the utility function.

How it's measured

Arrow-Pratt measure

The most common measure of risk aversion. It analyzes the utility representation to determine the degree of risk aversion.

Global risk aversion coefficients

Defined in terms of the average curvature of utility functions. They can be related to the slope of indifference curves.

Importance

Global risk aversion is a significant factor in business cycles and international equity correlations. Changes in US risk aversions can predict future real economic activity in emerging markets.

Q3. Describe attributes of Public goods?

Public goods have two main characteristics: non-rivalrous and non-excludable. These characteristics make it difficult for private firms to sell public goods.

Non-rivalrous

- The good doesn't disappear when more people use it
- One person's use of the good doesn't prevent others from using it
- For example, listening to a public radio broadcast doesn't reduce the availability of the broadcast for others

Non-excludable

- It's costly or impossible to prevent others from using the good
- Every citizen has access to the good
- For example, the US border patrol is available to all US residents at the same time

Examples of public goods

street lighting, clean water, parks, education, national defense, public sanitation systems, and storm sewers.